

Piecewise-Defined & Implicit Functions

ANSWER KEY



Evaluating piecewise functions

- 1 Given $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$, find $f(-3)$ and $f(4)$.

$f(-3)$: since $-3 < 0$, use x^2 : $f(-3) = (-3)^2 = 9$. $f(4)$: since $4 \geq 0$, use $2x + 1$: $f(4) = 2(4) + 1 = 9$.

- 2 A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.

$C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 5 + 1.5(w - 2)$ for $w > 2$. $C(5) = 5 + 1.5(5 - 2) = 5 + 4.5 = 9.5$, so 9.5 dollars.

Checking continuity of piecewise functions

- 3 Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.

For continuity, the two pieces must agree at $x = 2$: $3(2) + k = 2^2 - 1$, so $6 + k = 3$, giving $k = -3$.

- 4 Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

Left piece at $x = 1$: $1 + 1 = 2$. Right piece as $x \rightarrow 1^+$: $3(1) - 2 = 1$. Since $2 \neq 1$, the two pieces do not match at $x = 1$, so g is not continuous there — there is a jump discontinuity.

Visualizing a piecewise graph

- 5 The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.

a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there. Left piece as $x \rightarrow 0^-$: $-0 - 1 = -1$. Right piece at $x = 0$: $0^2 = 0$. Since $-1 \neq 0$, f has a jump discontinuity at $x = 0$, visible as the gap between the two branches in the graph.

b State the range of f shown on the graph. The linear branch for $x < 0$ takes all values greater than -1 (approaching but not including -1), and the parabola branch for $x \geq 0$ takes all values ≥ 0 . Combined range: $(-1, \infty)$.

Implicit relations and testing for functions

- 6 Determine whether $x^2 + y^2 = 25$ defines y as a function of x . Justify using the vertical line test.

This is a circle of radius 5. A vertical line such as $x = 3$ intersects it at two points ($y = 4$ and $y = -4$), so it fails the vertical line test — y is not a function of x .

- 7 Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.

$y = \pm\sqrt{25 - x^2}$. Each branch individually is a function: $y = \sqrt{25 - x^2}$ (upper semicircle) or $y = -\sqrt{25 - x^2}$ (lower semicircle) — together they form the full circle, which is not a function.

Implicit differentiation-style reasoning (algebraic slope)

- 8 For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y -values when $x = 2$, and describe what this means about the relation as a function.

$4 + 4y^2 = 16$, so $4y^2 = 12$, $y^2 = 3$, giving $y = \pm\sqrt{3}$. Since $x = 2$ produces two distinct y -values, the relation is not a function of x over its full domain.

Modeling with piecewise functions

- 9 A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x , and find the tax owed on an income of 25{,}000 dollars.

$T(x) = 0.10x$ for $0 \leq x \leq 10000$, and $T(x) = 1000 + 0.20(x - 10000)$ for $x > 10000$.
 $T(25000) = 1000 + 0.20(15000) = 1000 + 3000 = 4000$, so 4000 dollars in tax.

Piecewise functions with three or more pieces

- 10 A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.

$P(h) = 3$ for $0 < h \leq 1$; $P(h) = 3 + 2(h - 1)$ for $1 < h \leq 5$; $P(h) = 15$ for $h > 5$.
 $P(4) = 3 + 2(4 - 1) = 3 + 6 = 9$, so 9 dollars.

- 11 Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.

Since $2x - 6 = 0$ at $x = 3$: $f(x) = -(2x - 6) = 6 - 2x$ for $x < 3$, and $f(x) = 2x - 6$ for $x \geq 3$.
 $f(1) = 6 - 2(1) = 4$. $f(5) = 2(5) - 6 = 4$.

Graphing implicit relations point-by-point

- 12 For the implicit relation $y^2 = x + 4$, find all y -values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x -value.

At $x = 0$: $y^2 = 4$, so $y = \pm 2$. At $x = 5$: $y^2 = 9$, so $y = \pm 3$. At $x = -4$: $y^2 = 0$, so $y = 0$ (a single point). Since $y^2 \geq 0$ requires $x + 4 \geq 0$, the smallest allowed x -value is $x = -4$.

Piecewise functions and average rate of change

- 13 For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.

$$f(0) = 0^2 = 0 \text{ (using the first piece). } f(4) = 3(4) - 2 = 10 \text{ (using the second piece, since } 4 \geq 2\text{). AROC} \\ = \frac{10 - 0}{4 - 0} = \frac{10}{4} = 2.5.$$

- 14 Given $g(x) = 2x + 1$ for $x \leq 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

$$g(1) = 2(1) + 1 = 3 \text{ (first piece, since } 1 \leq 3\text{). } g(5) = 5^2 - 5 = 20 \text{ (second piece, since } 5 > 3\text{). AROC} \\ = \frac{20 - 3}{5 - 1} = \frac{17}{4} = 4.25.$$

- 15 A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

$M(x) = 20$ for $0 \leq x \leq 2$; $M(x) = 20 + 8(x - 2)$ for $2 < x \leq 8$ (since $20 + 8(6) = 68$ is reached exactly at $x = 8$); $M(x) = 68$ for $x > 8$. $M(8) = 20 + 8(8 - 2) = 20 + 48 = 68$, so 68 dollars.

Multiple choice — easy 15

- 16 Given $f(x) = 3x - 5$ for $x < 0$, and $f(x) = -5x + 9$ for $x \geq 0$, find $f(-5)$.

- a. 34
- b. -19
- c. -20 ✓
- d. -21

Since $-5 < 0$, use $3x - 5$: $f(-5) = -20$.

17 Given $f(x) = -2x - 5$ for $x < 3$, and $f(x) = 6x - 1$ for $x \geq 3$, find $f(2)$.

- a. 11
- b. -9 ✓
- c. -10
- d. -8

Since $2 < 3$, use $-2x - 5$: $f(2) = -9$.

18 Given $f(x) = -6x + 4$ for $x < 4$, and $f(x) = 6x + 5$ for $x \geq 4$, find $f(8)$.

- a. 53 ✓
- b. 54
- c. -44
- d. 52

Since $8 \geq 4$, use $6x + 5$: $f(8) = 53$.

19 Given $f(x) = x + 2$ for $x < -5$, and $f(x) = -5x - 9$ for $x \geq -5$, find $f(-2)$.

- a. 1 ✓
- b. -2
- c. 0
- d. 2

Since $-2 \geq -5$, use $-5x - 9$: $f(-2) = 1$.

20 Given $f(x) = -3x + 3$ for $x < 1$, and $f(x) = -6x - 9$ for $x \geq 1$, find $f(1)$.

- a. -15 ✓
- b. -14
- c. 0
- d. -16

Since $1 \geq 1$, use $-6x - 9$: $f(1) = -15$.

21 Given $f(x) = -5x + 1$ for $x < -1$, and $f(x) = 3x + 6$ for $x \geq -1$, find $f(-1)$.

a. 2

b. 3 ✓

c. 6

d. 4

Since $-1 \geq -1$, use $3x + 6$: $f(-1) = 3$.

22 Given $f(x) = 4x - 7$ for $x < 2$, and $f(x) = 6x - 5$ for $x \geq 2$, find $f(3)$.

a. 14

b. 5

c. 13 ✓

d. 12

Since $3 \geq 2$, use $6x - 5$: $f(3) = 13$.

23 Given $f(x) = 3x - 8$ for $x < -1$, and $f(x) = 2x$ for $x \geq -1$, find $f(-5)$.

a. -22

b. -24

c. -10

d. -23 ✓

Since $-5 < -1$, use $3x - 8$: $f(-5) = -23$.

24 Given $f(x) = 6x + 1$ for $x < -1$, and $f(x) = x - 2$ for $x \geq -1$, find $f(-5)$.

a. -29 ✓

b. -30

c. -28

d. -7

Since $-5 < -1$, use $6x + 1$: $f(-5) = -29$.

25 Given $f(x) = -5x + 5$ for $x < 5$, and $f(x) = 3x - 6$ for $x \geq 5$, find $f(6)$.

a. 12 ✓

b. 11

c. -25

d. 13

Since $6 \geq 5$, use $3x - 6$: $f(6) = 12$.

26 Given $f(x) = 2x - 7$ for $x < -1$, and $f(x) = -5x - 4$ for $x \geq -1$, find $f(-6)$.

a. 26

b. -18

c. -19 ✓

d. -20

Since $-6 < -1$, use $2x - 7$: $f(-6) = -19$.

27 Given $f(x) = x - 9$ for $x < 5$, and $f(x) = 4x - 9$ for $x \geq 5$, find $f(0)$.

a. -9 ✓

b. -6

c. -8

d. -10

Since $0 < 5$, use $x - 9$: $f(0) = -9$.

28 Given $f(x) = -3x + 8$ for $x < -3$, and $f(x) = -2x + 6$ for $x \geq -3$, find $f(-7)$.

a. 28

b. 30

c. 29 ✓

d. 20

Since $-7 < -3$, use $-3x + 8$: $f(-7) = 29$.

29 Given $f(x) = 5x + 2$ for $x < -4$, and $f(x) = -2x + 3$ for $x \geq -4$, find $f(-7)$.

a. -34

b. 17

c. -32

d. -33 ✓

Since $-7 < -4$, use $5x + 2$: $f(-7) = -33$.

30 Given $f(x) = -2x + 2$ for $x < 4$, and $f(x) = -x - 4$ for $x \geq 4$, find $f(-1)$.

a. 4 ✓

b. 5

c. -3

d. 3

Since $-1 < 4$, use $-2x + 2$: $f(-1) = 4$.

Multiple choice — medium 15

31 Find the value of k that makes $f(x) = x + k$ for $x < 4$, and $f(x) = -x - 1$ for $x \geq 4$, continuous at $x = 4$.

a. -10

b. -8

c. -9 ✓

d. -1

For continuity: $1(4) + k = -1(4) - 1$, so $4 + k = -5$, giving $k = -9$.

32 Find the value of k that makes $f(x) = -2x + k$ for $x < 2$, and $f(x) = 7x - 1$ for $x \geq 2$, continuous at $x = 2$.

a. 17 ✓

b. 16

c. 9

d. 18

For continuity: $-2(2) + k = 7(2) - 1$, so $-4 + k = 13$, giving $k = 17$.

33 Find the value of k that makes $f(x) = -8x + k$ for $x < 5$, and $f(x) = 5x + 1$ for $x \geq 5$, continuous at $x = 5$.

a. 67

b. -14

c. 66 ✓

d. 65

For continuity: $-8(5) + k = 5(5) + 1$, so $-40 + k = 26$, giving $k = 66$.

- 34 Find the value of k that makes $f(x) = -9x + k$ for $x < -2$, and $f(x) = -7x + 6$ for $x \geq -2$, continuous at $x = -2$.
- a. 38
 - b. 2 ✓
 - c. 3
 - d. 1

For continuity: $-9(-2) + k = -7(-2) + 6$, so $18 + k = 20$, giving $k = 2$.

- 35 Find the value of k that makes $f(x) = 7x + k$ for $x < 2$, and $f(x) = -5x - 8$ for $x \geq 2$, continuous at $x = 2$.
- a. -32 ✓
 - b. -31
 - c. -4
 - d. -33

For continuity: $7(2) + k = -5(2) - 8$, so $14 + k = -18$, giving $k = -32$.

- 36 Find the value of k that makes $f(x) = 3x + k$ for $x < 6$, and $f(x) = -5x - 8$ for $x \geq 6$, continuous at $x = 6$.
- a. -57
 - b. -20
 - c. -55
 - d. -56 ✓

For continuity: $3(6) + k = -5(6) - 8$, so $18 + k = -38$, giving $k = -56$.

- 37 Find the value of k that makes $f(x) = 3x + k$ for $x < -2$, and $f(x) = x - 9$ for $x \geq -2$, continuous at $x = -2$.

a. -5 ✓

b. -6

c. -17

d. -4

For continuity: $3(-2) + k = 1(-2) - 9$, so $-6 + k = -11$, giving $k = -5$.

- 38 Find the value of k that makes $f(x) = -9x + k$ for $x < -4$, and $f(x) = 2x - 6$ for $x \geq -4$, continuous at $x = -4$.

a. -49

b. 22

c. -51

d. -50 ✓

For continuity: $-9(-4) + k = 2(-4) - 6$, so $36 + k = -14$, giving $k = -50$.

- 39 Find the value of k that makes $f(x) = x + k$ for $x < 3$, and $f(x) = x - 5$ for $x \geq 3$, continuous at $x = 3$.

a. -4

b. -5 ✓

c. -6

d. 1

For continuity: $1(3) + k = 1(3) - 5$, so $3 + k = -2$, giving $k = -5$.

- 40 Find the value of k that makes $f(x) = -8x + k$ for $x < 5$, and $f(x) = -2x + 2$ for $x \geq 5$, continuous at $x = 5$.

- a. 33
- b. -48
- c. 31
- d. 32 ✓

For continuity: $-8(5) + k = -2(5) + 2$, so $-40 + k = -8$, giving $k = 32$.

- 41 Find the value of k that makes $f(x) = 7x + k$ for $x < 0$, and $f(x) = 8x - 9$ for $x \geq 0$, continuous at $x = 0$.

- a. -7
- b. -9 ✓
- c. -8
- d. -10

For continuity: $7(0) + k = 8(0) - 9$, so $0 + k = -9$, giving $k = -9$.

- 42 Find the value of k that makes $f(x) = 9x + k$ for $x < 3$, and $f(x) = -x - 2$ for $x \geq 3$, continuous at $x = 3$.

- a. -33
- b. -31
- c. 22
- d. -32 ✓

For continuity: $9(3) + k = -1(3) - 2$, so $27 + k = -5$, giving $k = -32$.

43 Find the value of k that makes $f(x) = -7x + k$ for $x < 3$, and $f(x) = 7x - 8$ for $x \geq 3$, continuous at $x = 3$.

- a. 35
- b. -8
- c. 33
- d. 34 ✓

For continuity: $-7(3) + k = 7(3) - 8$, so $-21 + k = 13$, giving $k = 34$.

44 Find the value of k that makes $f(x) = x + k$ for $x < -1$, and $f(x) = 8x + 3$ for $x \geq -1$, continuous at $x = -1$.

- a. -4 ✓
- b. -6
- c. -5
- d. -3

For continuity: $1(-1) + k = 8(-1) + 3$, so $-1 + k = -5$, giving $k = -4$.

45 Find the value of k that makes $f(x) = 8x + k$ for $x < 5$, and $f(x) = -x + 5$ for $x \geq 5$, continuous at $x = 5$.

- a. 40
- b. -41
- c. -39
- d. -40 ✓

For continuity: $8(5) + k = -1(5) + 5$, so $40 + k = 0$, giving $k = -40$.

Multiple choice — hard 15

46 For the implicit relation $x^2 + y^2 = 625$, find all y -values when $x = 24$.

a. $y = \pm 25$

b. $y = \pm 24$

c. $y = \pm 7$ ✓

d. $y = 7$

$24^2 + y^2 = 625$: $y^2 = 625 - 576 = 49$, so $y = \pm 7$.

47 For the implicit relation $x^2 + y^2 = 289$, find all y -values when $x = 15$.

a. $y = 8$

b. $y = \pm 8$ ✓

c. $y = \pm 17$

d. $y = \pm 15$

$15^2 + y^2 = 289$: $y^2 = 289 - 225 = 64$, so $y = \pm 8$.

48 For the implicit relation $x^2 + y^2 = 25$, find all y -values when $x = 3$.

a. $y = \pm 4$ ✓

b. $y = 4$

c. $y = \pm 3$

d. $y = \pm 5$

$3^2 + y^2 = 25$: $y^2 = 25 - 9 = 16$, so $y = \pm 4$.

49 For the implicit relation $x^2 + y^2 = 841$, find all y -values when $x = 21$.

a. $y = 20$

b. $y = \pm 20$ ✓

c. $y = \pm 21$

d. $y = \pm 29$

$21^2 + y^2 = 841$: $y^2 = 841 - 441 = 400$, so $y = \pm 20$.

50 For the implicit relation $x^2 + y^2 = 1681$, find all y -values when $x = 9$.

a. $y = 40$

b. $y = \pm 40$ ✓

c. $y = \pm 41$

d. $y = \pm 9$

$9^2 + y^2 = 1681$: $y^2 = 1681 - 81 = 1600$, so $y = \pm 40$.

51 For $f(x) = x + 7$ on $x < 1$ and $f(x) = -4x + 2$ on $x \geq 1$, find the average rate of change of f on $[-3, 7]$.

a. -4

b. -3 ✓

c. 3

d. -2

$f(-3) = 4$ (first piece). $f(7) = -26$ (second piece, since $7 \geq 1$). $\text{AROC} = \frac{-26 - (4)}{7 - (-3)} = -3$.

52 For $f(x) = 5x - 8$ on $x < 0$ and $f(x) = -3x + 7$ on $x \geq 0$, find the average rate of change of f on $[-1, 2]$.

a. $\frac{14}{3}$ ✓

b. $-\frac{14}{3}$

c. $\frac{11}{3}$

d. $\frac{17}{3}$

$f(-1) = -13$ (first piece). $f(2) = 1$ (second piece, since $2 \geq 0$). $\text{AROC} = \frac{1 - (-13)}{2 - (-1)} = \frac{14}{3}$.

53 For $f(x) = -5x - 8$ on $x < 2$ and $f(x) = 4x + 7$ on $x \geq 2$, find the average rate of change of f on $[-3, 7]$.

a. $-\frac{14}{5}$

b. $\frac{9}{5}$

c. $\frac{14}{5}$ ✓

d. $\frac{19}{5}$

$f(-3) = 7$ (first piece). $f(7) = 35$ (second piece, since $7 \geq 2$). $\text{AROC} = \frac{35 - (7)}{7 - (-3)} = \frac{14}{5}$.

54 For $f(x) = -x + 8$ on $x < -3$ and $f(x) = 6x + 7$ on $x \geq -3$, find the average rate of change of f on $[-5, 0]$.

a. $-\frac{6}{5}$ ✓

b. $-\frac{11}{5}$

c. $-\frac{1}{5}$

d. $\frac{6}{5}$

$f(-5) = 13$ (first piece). $f(0) = 7$ (second piece, since $0 \geq -3$). $\text{AROC} = \frac{7 - (13)}{0 - (-5)} = -\frac{6}{5}$.

55 For $f(x) = 4x - 3$ on $x < -1$ and $f(x) = x - 6$ on $x \geq -1$, find the average rate of change of f on $[-2, 0]$.

a. $\frac{7}{2}$

b. $\frac{5}{2}$ ✓

c. $-\frac{5}{2}$

d. $\frac{3}{2}$

$f(-2) = -11$ (first piece). $f(0) = -6$ (second piece, since $0 \geq -1$). $\text{AROC} = \frac{-6 - (-11)}{0 - (-2)} = \frac{5}{2}$.

- 56 A shipping company charges a flat 7 dollars for packages up to 3 kg, plus $\frac{5}{2}$ dollars per kg for anything over 3 kg. Find the cost for a 13 kg package.

a. 32 ✓

b. 33

c. 31

d. $\frac{79}{2}$

$$C(13) = 7 + \frac{5}{2}(13 - 3) = 7 + \frac{5}{2}(10) = 32 \text{ dollars.}$$

- 57 A shipping company charges a flat 8 dollars for packages up to 3 kg, plus 3 dollars per kg for anything over 3 kg. Find the cost for a 11 kg package.

a. 33

b. 41

c. 32 ✓

d. 31

$$C(11) = 8 + 3(11 - 3) = 8 + 3(8) = 32 \text{ dollars.}$$

- 58 A shipping company charges a flat 13 dollars for packages up to 2 kg, plus 2 dollars per kg for anything over 2 kg. Find the cost for a 7 kg package.

a. 27

b. 24

c. 23 ✓

d. 22

$$C(7) = 13 + 2(7 - 2) = 13 + 2(5) = 23 \text{ dollars.}$$

- 59 A shipping company charges a flat 14 dollars for packages up to 1 kg, plus $\frac{5}{2}$ dollars per kg for anything over 1 kg. Find the cost for a 2 kg package.

a. $\frac{33}{2}$ ✓

b. 19

c. $\frac{35}{2}$

d. $\frac{31}{2}$

$$C(2) = 14 + \frac{5}{2}(2 - 1) = 14 + \frac{5}{2}(1) = \frac{33}{2} \text{ dollars.}$$

- 60 A shipping company charges a flat 8 dollars for packages up to 2 kg, plus 3 dollars per kg for anything over 2 kg. Find the cost for a 9 kg package.

a. 29 ✓

b. 30

c. 28

d. 35

$$C(9) = 8 + 3(9 - 2) = 8 + 3(7) = 29 \text{ dollars.}$$